Response to Reviewer 1

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Discrete Symplectic Cosmology: a phenomenological lattice surrogate for CMB-like fluctuations

May 23, 2026

We thank Reviewer 1 for an unusually constructive and detailed report. The eight major comments and the minor remarks have substantially shaped the revision: we have retracted several quantitative claims, replaced the toy analytic reference with a CAMB Planck-2018 benchmark, made the symplecticity caveat explicit, demoted the Big-Bang-to-cosmic-web section to a phenomenological visualization, and promoted the held-out hemisphere result from a footnote-level caveat to one of the main findings. The revised abstract and introduction now position the work explicitly as an exploratory toy-model study of a differentiable lattice surrogate, in line with the reviewer's framing.

In the response below, the reviewer's comments are quoted in italics; our reply, the resulting changes, and pointers to the revised manuscript follow each item.

Major comments

Comment 1 — Symplecticity is broken by the drag term

> *First, I am not fully convinced that the simulations used in the main phenomenological comparisons are genuinely symplectic, in the relevant sense. The manuscript introduces a drag term, and it is explicitly shown that the phase-space volume contracts when $\eta \neq 0$. This appears difficult to reconcile with the repeated emphasis on symplecticity as the central physical principle. If the intended framework is instead a weakly damped or conformally symplectic system, this should be formulated and interpreted accordingly.*

We agree. The integrator is exactly symplectic only when $\eta = 0$, and the runs that combine the lattice with the inverse reconstruction or the late-time continuation use $\eta \in [0.01, 0.06]$, which contracts phase-space volume. We have made this distinction explicit and adopted the reviewer's terminology of *weakly damped symplectic* (conformally symplectic) for the runs with $\eta > 0$.

Changes:

- Section 3.1 (*Two-dimensional weakly damped symplectic lattice*) now opens with an explicit statement that the integrator is exactly symplectic only at $\eta = 0$, identifies which sections use $\eta = 0$ (Gaussianity ensemble and acoustic-peak runs) and which use $\eta > 0$ (Phase I cosmic timeline; inverse reconstruction), and states that quantitative claims depending on the symplecticity should be read as "weakly damped symplectic" rather than "exactly symplectic".

- The abstract and introduction now describe the lattice as "a non-autonomous, weakly damped symplectic map".
- In the Discussion (Section 5), the paragraph that previously justified Gaussianity through phase-space volume preservation has been replaced by a more conservative statement (linear regime + central limit theorem; symplecticity not isolated by the present ablations).

Comment 2 — Acoustic freeze-out is not a finite horizon

Second, the acoustic freeze-out argument appears problematic. The manuscript notes that the formal horizon integral with $c_s \sim 1/\ln n$ diverges, but elsewhere the discussion seems to treat the acoustic horizon as effectively finite in a physically analogous way to recombination. As far as I can tell, the reported finite scale is threshold- and simulation-dependent, rather than a true future sound horizon. This weakens one of the central claimed analogies with standard CMB physics.

We agree, and we have rewritten the freeze-out section to be explicit about this. The formal integral $\int_0^\infty c_{\text{eff}}(t)\, dt$ with $c_{\text{eff}} \sim 1/\ln^2(t)$ diverges, and the value $r_s \approx 38$ that we report is a threshold-defined effective scale.

Changes:

- Section 4.4 has been retitled *Threshold-defined effective freeze-out scale.*
- A new paragraph spells out the logarithmic divergence of the horizon integral and identifies three sources of dependence: amplitude threshold, lattice size, and integration length.
- A new Table (*Sensitivity of r_s to threshold and grid size*) reports r_s for percentile thresholds $\in \{90\%, 95\%, 99\%\}$ and lattice sizes $\in \{150, 300, 600\}$, showing the expected drift.
- The Discussion no longer claims r_s as an analog of the comoving sound horizon.

Comment 3 — The Gaussianity result is weaker than claimed

Third, I find the interpretation of Gaussianity overstated. If the initial conditions are already nearly Gaussian and the symplectic evolution is mostly linear, near-Gaussian one-point statistics is not unexpected. The comparison with a dissipative coupled-map lattice is useful, but it does not isolate symplecticity alone, since the heuristic differs in several other ways. I am therefore not convinced that the paper demonstrates a nontrivial emergence of CMB-like Gaussianity from the proposed dynamics.

We agree. The dynamics are approximately linear in the parameter range we explore, and the initial conditions are themselves Gaussian, so near-Gaussian one-point statistics is the expected outcome rather than evidence for the proposed dynamics.

Changes:

- Section 4.1 has been retitled *Near-Gaussian statistics: a consistency check, not an emergence claim.*
- The main paragraph now states explicitly: "We do not interpret this as a non-trivial emergence of CMB-like Gaussianity. The initial conditions are random Gaussian fields, the Störmer–Verlet update is linear in ϕ in the explored parameter range, and a linear evolution of a Gaussian field is also Gaussian."
- Section 4.2 (the dissipative-CML ablation) now explicitly cautions that the comparison cannot isolate symplecticity, because the two baselines differ simultaneously in nonlinearity, conservation properties, integration order, and stencil.
- The abstract has been edited to say that near-Gaussian one-point statistics are "consistent with expectations rather than a non-trivial emergence".

Comment 4 — Comparison with the real CMB is not strong enough

> *Fourth, the comparison with the real CMB is not sufficiently strong. The most favorable spectral comparisons are made against a toy analytic reference spectrum, not against CAMB and CLASS. When the model is compared with the (Planck) angular spectrum, the manuscript reports reduced chi-square values substantially larger than unity and notes that the acoustic peak structure is not reproduced in the projected angular spectrum. This seems to me a round result, and it should substantially temper the claim that the model reproduces key CMB features.*

We have substantially tempered the claim and added a real CAMB benchmark.

Changes:

- A new notebook `notebooks/14_camb_benchmark.ipynb` computes the Planck-2018 best-fit D_ℓ^{TT} via CAMB ($H_0 = 67.32$, $\Omega_b h^2 = 0.02238$, $\Omega_c h^2 = 0.1201$, $A_s = 2.1 \times 10^{-9}$, $n_s = 0.9649$, $\tau = 0.0543$) and compares it to the DSC sky-map spectrum from a 5-seed ensemble.
- Section 4.10 has been retitled *Quantitative comparison with the CAMB Planck-2018* C_ℓ and rewritten around this benchmark. With a single overall amplitude as the only fit parameter and a generous 20% per-ℓ error budget, we obtain $\chi^2/\mathrm{dof} \approx 80$ over $\ell \in [30, 256]$ and $\chi^2/\mathrm{dof} \approx 4.6$ over the peak band $\ell \in [150, 256]$.
- A new Table (χ^2 *benchmark vs. CAMB*) reports sensitivity to the projection radius R_{shell} and lattice size N_{grid}, confirming that the agreement depends on geometric choices, not on intrinsic physics.
- A new Figure 17 (`fig17_camb_benchmark.pdf`) visualizes the comparison.
- The abstract, introduction, conclusion, and discussion all now state explicitly that we retract any quantitative claim of reproducing the Planck C_ℓ at the level of individual multipoles.

Comment 5 — Lattice k-peaks may be standing-wave artifacts

Fifth, the "acoustic peaks" in the lattice spectrum may simply be standing-wave features of the finite lattice and boundary conditions. While this is mathematically interesting, I do not see a demonstrated physical mapping between the lattice k-space peaks and the observed CMB multipoles. Without such a mapping, the analogy with baryon-photon acoustic oscillations remains qualitative.

We agree, and we have made the geometric mapping explicit and quantitative.

Changes:

- In Section 4.10 (and in the new notebook 14) we use the flat-sky / Limber projection $\ell \approx 2\pi R_{\text{shell}}\, k$ to translate lattice peaks to multipoles. With our nominal $R_{\text{shell}} = 0.35$, the first three lattice $D(k)$ peaks at $k \approx 13, 28, 42$ map to $\ell \approx 29, 62, 92$, far from Planck's $\ell \approx 220, 540, 810$.
- Section 4.10 also reports a sensitivity scan over R_{shell} and N_{grid} showing that the lattice peak positions in ℓ-space drift in a manner consistent with the standing-wave hypothesis.
- The Discussion paragraph on acoustic peaks now states: "We therefore retract the earlier framing of these peaks as analogs of baryon–photon acoustic oscillations."

Comment 6 — Riemann zero / cosmology connection is conjectural

Sixth, the connection between Riemann-zeta spectral statistics and cosmological evolution remains highly conjectural. I would not object to such a speculative idea in principle, but in that case the numerical evidence must be quantitatively much more compelling.

We have demoted the Riemann connection from a derivation to an inspiration / working hypothesis.

Changes:

- The introduction now explicitly states: "We stress that the program of identifying a physical Hamiltonian whose spectrum reproduces the Riemann zeros remains a conjecture, and the present work uses this connection only as a working motivation for a particular functional form, not as an established physical correspondence."
- The phrase "derived from Riemann zero spectral statistics" has been replaced by "inspired by" (abstract, introduction, conclusion).
- The $1/\ln^2$ functional form is now described as "a definite ansatz to study" rather than as a derived prediction.
- The Discussion section explicitly states that the present work treats the Riemann-zero connection as a working motivation for the $1/\ln^2$ cooling-law functional form only, and explicitly does *not* engage with the broader Hilbert–Pólya program or with experimental programs aimed at realizing the Riemann spectrum in laboratory systems; those questions are outside the scope of the lattice cosmology presented here.

Comment 7 — Inverse reconstruction is highly degenerate; the held-out test is the diagnostic

> *Seventh, the inverse reconstruction section demonstrates differentiability, but not physical reconstruction. The high training correlation with Planck pixels is not surprising given the very large number of free 3D parameters. The held-out hemisphere test, which folds in any case, is the more important diagnostic. This result suggests that the reconstruction is moderate at best and should not be interpreted as evidence for a physically meaningful inferred 3D primordial field.*

We agree. The held-out hemisphere result $r = -0.17$ is now treated as the primary diagnostic for the inverse reconstruction.

Changes:

- In the introduction, the held-out hemisphere result has been moved to position 1 of the main findings list, ahead of all other results.
- The abstract now states the in-sample $r = 0.98$ and held-out $r = -0.17$ side by side, and concludes: "the recovered 3D field cannot be interpreted as a physical primordial reconstruction".
- The Discussion devotes a paragraph to forward identifiability vs. field-level underdetermination, explicitly comparing $\sim 10^6$ unknowns to $\sim 10^4$ pixels.
- The conclusion now positions the differentiable engine as a methodological building block (clean gradients through 25 Störmer–Verlet steps and the sphere-projection layer) rather than as a tool for inferring the 3D primordial state from the CMB alone.

Comment 8 — "Big Bang to cosmic web" is too speculative

> *Finally, I find the "Big Bang to cosmic web" section too speculative. The Allen-Cahn-type nonlinearity used for late-time structure formation is not a standard gravitational model. The resulting visual resemblance to a cosmic web is interesting, but I do not think it supports the cosmological interpretation currently attached to it.*

We agree, and we have demoted this section to a phenomenological visualization.

Changes:

- Section 4.12 has been retitled *Phenomenological visualization: continuing the lattice past the CMB epoch.*
- The opening paragraph now states explicitly: "This subsection presents a phenomenological visualization of the lattice continued beyond the CMB epoch, not a model of cosmic structure formation… the Allen–Cahn nonlinearity used here is not gravity."
- The Allen–Cahn nonlinearity is described as "a generic toy potential, not a gravitational interaction".
- The earlier analogy with cosmic inflation is explicitly retracted, both in this section and in the corresponding paragraph of the Discussion.

- The figure caption (Figure 16) now states: "Visual resemblance to the cosmic web is generic to domain coarsening; no quantitative match to large-scale structure is claimed."

Closing remark

The combined effect of these changes is a significantly more conservative paper: claims that we cannot quantitatively support have been retracted, the symplectic-vs-weakly-damped distinction is now explicit, the Riemann-zero motivation is presented as a working hypothesis rather than a derivation, and the Planck C_ℓ benchmark is now done against CAMB rather than against a toy analytic reference. The differentiable lattice surrogate, the held-out hemisphere diagnostic, and the explicit standing-wave interpretation of the lattice peaks are, we believe, useful methodological contributions even after the more speculative claims have been removed.

Beyond Reviewer 1's comments: cross-scale empirical context

The eight points above directly address the reviewer's report. In addition, we have used this revision opportunity to consolidate, in a single manuscript, four further empirical pillars of the DSC framework that previously lived in companion preprints (Wang, *Cosmological Evolution as a Non-autonomous Dynamical System*, Zenodo `10.5281/zenodo.19218674`; Wang, *Discrete Symplectic Cosmology*, Zenodo `10.5281/zenodo.19429778`). Reviewer 1's comment 6 noted that the connection between Riemann-zeta spectral statistics and cosmological evolution remains highly conjectural, and asked that the numerical evidence be made quantitatively more compelling for any speculative claim. We agree, and the new §4.13 (*Cross-scale empirical pillars beyond the CMB lattice*) presents the full evidence portfolio rather than only the CMB lattice slice. The headline content of the new section is:

- **Pillar A.** A 128-absorber subset analysis of quasar $\Delta\alpha/\alpha$ data fits the DSC $1/\ln^2(t)$ drift with $\Delta\chi^2 = 31.5$ improvement over the constant null at one extra parameter (Notebook 15).
- **Pillar B.** The same comparison on the full King *et al.* (2012) 295-absorber catalog (CDS J/MNRAS/422/3370) yields $\Delta\chi^2 \approx 0$ and $\Delta\mathrm{BIC} = +5.7$ in favor of the null, with joint clock+quasar bound $|\Gamma| < 0.30$ (Notebook 17). The subset tension between Pillars A and B is reported explicitly rather than glossed.
- **Pillar C.** A four-anchor H_0 fit (SH0ES, JWST, Planck, BAO) by $H(t) = H_\infty + \beta/\ln^2(t/t_P)$ achieves $R^2 = 0.91$, reduced $\chi^2 = 0.28$ (Notebook 16), recasting the Hubble tension as a smooth time-dependent relaxation. This is the strongest single piece of empirical support for the framework that we currently have.
- **Pillar D.** A CAMB + DESI BAO Fisher analysis gives $\sigma(\gamma_{\mathrm{eff}}) \sim 1.2 \times 10^4$ (Notebook 18), so the DSC dark-energy correction is unconstrained at present sensitivity; this is consistent with the negative CMB lattice benchmark of §4.10.

- **Pillar E.** Five late-time stability criteria pass for $\gamma_{\text{eff}} \in [0, 10]$ over $a \in [10^{-10}, 1]$ (Notebook 19).

Why this material now lives in the main manuscript. We had originally split the empirical evidence (Pillars A–E) and the methodological CMB lattice surrogate (the original manuscript) into two preprints (Zenodo `19218674` and `19429778`) for length reasons. The reviewer's comments make clear that, evaluated on the CMB lattice slice alone, the framework's quantitative case is weak. We therefore consolidate the empirical pillars into the same paper, so that the framework can be evaluated on the totality of the cross-scale evidence rather than on any single observational probe. Of the five pillars, two are positive (Pillars C and E), two are null at current sensitivity (Pillars B and D), and one is suggestive but inconclusive (Pillar A). We do not claim the cross-scale evidence *establishes* the framework; we claim only that no current observation *rules it out*, and that the JWST H_0 pillar provides a useful $1/\ln^2$ scaling that connects to the standing Hubble tension.

New material added in this revision (relative to the version reviewed):

- Five Pillar A–E notebooks: `notebooks/15_alpha_drift_quasar.ipynb`, `16_jwst_h0_anchors.ipy` `17_quasar_293_fit.ipynb`, `18_camb_fisher.ipynb`, `19_late_time_stability.ipynb`.
- Five further Pillar F–K notebooks added to give compelling numerical evidence at high redshift resolution: `20_pillar_F_cosmic_chronometers.ipynb` (32 cosmic-chronometer points), `21_pillar_G_pantheon.ipynb` (Pantheon+ 1580 SNe with full covariance), `22_pillar_H_desi_dr2.ipynb` (DESI DR2 BAO 11 measurements), `23_pillar_J_join` (joint 1623-constraint analysis with ΛCDM/CPL/DSC), `24_pillar_K_subset_scan.ipynb` (King 2012 subset decomposition).
- Ten new figures: `fig18_alpha_drift_quasar.pdf` through `fig27_pillar_K_subset.pdf`.
- Two new sections: §4.13 *Cross-scale empirical pillars beyond the CMB lattice* (Pillars A–E) and §4.14 *Late-universe falsification: Pillars F–K with high-precision data.* The latter reports the late-universe falsifications of the simple DSC ansatz against cosmic chronometers ($\chi^2/\text{dof} = 4.0$ vs ΛCDM's 0.49), Pantheon+ ($\Delta\chi^2 = +617$ in favor of ΛCDM, $\sim 25\sigma$-equivalent), DESI DR2 BAO ($\Delta\chi^2 = +14000$), their joint combination ($\Delta\chi^2 = +37{,}000$, while CPL is mildly preferred over ΛCDM at $\Delta\text{BIC} = -2.6$), and a cut-by-cut decomposition of the King 2012 quasar catalog showing that Pillar A's apparent signal is the long-standing Keck α-dipole.
- Abstract, introduction main-findings list, and conclusion are now organized around eleven pillars and explicitly retract the claims that DSC fits late-time $H(z)$, BAO, supernovae, or the Planck C_ℓ competitively with ΛCDM. We retract any quantitative claim that the simple DSC ansatz is a viable cosmological model on dense late-time data.
- References `Rosenband2008`, `King2012`, `RiessSH0ES2022`, `RiessJWST2024`, `Moresco2022`, `Scolnic2022`, `Brout2022`, `DESI2025DR2`, `Whitmore2015`, `Wilczynska2020` added to `references.bib`.

We are grateful to the reviewer for the level of detail in the report, which directly enabled this rewrite.